



NATHAN RAWIRI

MECHANICAL & AEROSPACE ENGINEER

CONTACT

+61 433 963 608

rawirinathan@gmail.com

EDUCATION

2021-2025

UNIVERSITY OF QUEENSLAND

- BE(Hons) Mechanical and Aerospace Engineering Degree
- GPA 5.6

2024 Semester Exchange

TU DELFT, NETHERLANDS

- Aerospace Engineering

2019-2020

UNIVERSITY OF OTAGO

- Computer Science & Economics

2018

UNIVERSITY OF CANTERBURY

- 1st Year General Engineering

SOFT SKILLS

- Communication
- Project Management
- Teamwork
- Social
- Problem Solving - MENSA IQ
- Critical Thinking
- Emotional Intelligence
- NZSIA Snowboard Instructor
- Half-Ironman Finisher (2023)

CASUAL WORK

- Fishbowl - Hospitality
- O'Brian Group - Hospitality
- Uber Eats - Delivery Biker
- Fulton Hogan - Road Repairs
- Domino's - Delivery Driver

PORTFOLIO

<https://rawnatural.github.io>

TECHNICAL SKILLS

- **CAD and Simulation:** Siemens NX, Inventor, Simulink, FEA, CFD
- **Manufacturing:** Lathe, Mill, Press Brake, CNC, Laser Cutter, Bandsaw
- **Software:** Python, PHP, MySQL, C++, Solidity (blockchain), iOS Swift

WORK EXPERIENCE

Siemens Mobility

2023

Project Engineer

- Managed Complex \$50m Railway Signalling Project in Wellington, NZ
- Replaced mechanical interlocking system with Westrace MkII & ETCS.
- Generated installation location requirements for 168 Axle Counters.
- Aligned bill of materials with designs and procurement.
- Procured \$millions of signalling equipment before long lead-times.
- Chaired meetings and organised professional engineering team.
- Lead Project Engineer for close-out of post-commisioned project.

UQ Space

2021- 2025

Aerostructures Engineer

- Develop Rockets and Rovers, representing UQ at competitions.
- Lead Manufacturer of composite airframes and aluminium rover.
- Fibreglass/epoxy layups vacuum bagged on mandrels and moulds.
- 1st place in 5000ft category at AURC 2025.
- Deputy Project Lead for 2nd place in Australian Rover Challenge 2025
- Extra-Curricular professional engineering teamwork experience.

Software Developer

2019-Present

Self Employed

- Full-Stack website development - PHP, JS, HTML, CSS, MySQL
- Engineering Solution Optimisation - Python, C++
- Blockchain development - Solidity smart contracts - web3 integration
- iOS app development - Swift
- <https://github.com/RawNatural>

Manufacturing Supervisor

2025

UQ Innovate

- Innovate is the makerspace facility at UQ equipped with advanced manufacturing tools such as lathes, mills, 3D printers and laser cutters
- Offer technical support and ensure safe operation of machinery/tools

REFERENCE

Mark Kilby

Siemens Head of Project, Systems and Specialty Engineering

Phone: +64 22 017 2014

Email: mark.kilby@siemens.com

Oliver Pettingill

UQ Space Aerostructures Lead
Mercedes AMG Powertrain Engineer

Email :

Oliver.pettingill01@mercedes-amg-hpp.com